

CS492D: Diffusion Models and Their Applications

Denoising Diffusion Implicit Models 2 / Classifier-Free Guidance

LECTURE 6
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Fall 2024
KAIST

RECAP: DDPM vs. DDIM

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0, \left(\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t\right) \mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. Compute $\mathbf{x}_{0|t} = \frac{1}{\sqrt{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t))$
2. Compute $\tilde{\boldsymbol{\mu}} = \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_{0|t}$.
3. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sqrt{\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}} \beta_t \mathbf{z}_t$.

RECAP: DDPM vs. DDIM

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

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4. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sigma_t \mathbf{z}_t.$

RECAP: DDPM vs. DDIM

Q. What is $\tilde{\mu}$ with respect to x_t and ϵ_t for both DDPM and DDIM cases?

RECAP: DDPM vs. DDIM

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}}\boldsymbol{\varepsilon}_t\right), \left(\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t\right)\mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

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RECAP: DDPM vs. DDIM

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\tilde{\boldsymbol{\mu}} = \sqrt{\bar{\alpha}_{t-1}}\left(\frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}_t}{\sqrt{\bar{\alpha}_t}}\right) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \boldsymbol{\epsilon}_t, \sigma_t^2 \mathbf{I}\right)$$

For each time step $t = T, \dots, 1$, repeat:

1. Compute $\tilde{\boldsymbol{\mu}} = \sqrt{\bar{\alpha}_{t-1}}\left(\frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}}\right) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$.
2. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
3. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \sigma_t \mathbf{z}_t$.

RECAP: DDIM

Q. Check out that DDPM is a special case of DDIM when

$$\sigma_t^2 = \tilde{\sigma}_t^2 = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t.$$

RECAP: DDIM

Controlling Stochasticity

$$\text{Let } \sigma_t = \eta \sqrt{\frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}} \beta_t.$$

- $\eta = 0$: **Deterministic** process.

For the same $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, we always obtain the same $\mathbf{x}_0$.

- $\eta = 1$: Same as **DDPM**.

RECAP: DDIM

Q. What is the maximum σ_t ?

RECAP: DDIM

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\tilde{\mu} = \sqrt{\bar{\alpha}_{t-1}} \left(\frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_t}{\sqrt{\bar{\alpha}_t}} \right) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \boldsymbol{\epsilon}_t, \sigma_t^2 \mathbf{I}\right)$$

A. $1 - \bar{\alpha}_{t-1} - \sigma_t^2$ must be non-negative.

The maximum σ_t is $\sqrt{1 - \bar{\alpha}_{t-1}}$.

RECAP: Accelerating Sampling Process

The reverse process for a **sub-sequence** time steps $\tau = [\tau_1, \tau_2, \dots, \tau_S]$:

$$p_{\theta}(\mathbf{x}_{\tau}) = p(\mathbf{x}_T) \prod_{t=1}^S p_{\theta}(\mathbf{x}_{\tau_{i-1}} | \mathbf{x}_{\tau_i})$$

RECAP: Accelerating Sampling Process

- As **smaller** time steps are used, the quality of the generated data can worsen.
- However, quality degradation is **mitigated** when the DDIM reverse process becomes more **deterministic**.

RECAP: DDIM Summary

- DDPM is a special case of DDIM.
- No need to retrain the noise predictor.
- The reverse process becomes deterministic when $\sigma_t = 0$.
- Quality degradation with fewer time steps is mitigated when σ_t is small.

RECAP: Accelerating Sampling Process

- When $\eta = 0$ (deterministic),
- the quality does not get bad too much even when $S = 10$.
- The quality when $S = 1000$ is even better than DDPM!

S	CIFAR10 (32×32)					CelebA (64×64)					
	10	20	50	100	1000	10	20	50	100	1000	
η	0.0	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98

Assignment 2

- Assignment 2 is about changing DDPM to DDIM.
- It builds on your solution from Assignment 1. You can start Assignment 2 when you finish Assignment 1!

Classifier-Free Guidance (CFG)

Ho and Salimans, Classifier-Free Diffusion Guidance, NeurIPS 2021 Workshop.

Classifier-Free Guidance (CFG)

How to make a **conditional** diffusion model?

Dog →



Classifier-Free Guidance (CFG)

Modify the noise prediction network to take the condition (label) y as an additional input:

$$\hat{\epsilon}_{\theta}(\mathbf{x}_t, t) \rightarrow \hat{\epsilon}_{\theta}(\mathbf{x}_t, \mathbf{y}, t)$$

Classifier-Free Guidance (CFG)

We can jointly train the **conditional** and **unconditional** diffusion models by introducing a null label ϕ .

$\hat{\epsilon}_{\theta}(\mathbf{x}_t, \phi, t)$: Noise prediction *without* condition.

Classifier-Free Guidance (CFG)

How can we **enhance** the conditioning?

We can extrapolate the conditional noise $\hat{\epsilon}_{\theta}(\mathbf{x}_t, \mathbf{y}, t)$ from the null-condition noise $\hat{\epsilon}_{\theta}(\mathbf{x}_t, \phi, t)$:

$$\tilde{\epsilon}_{\theta}(\mathbf{x}_t, \mathbf{y}, t) = (1 + \omega)\hat{\epsilon}_{\theta}(\mathbf{x}_t, \mathbf{y}, t) - \omega\hat{\epsilon}_{\theta}(\mathbf{x}_t, \phi, t)$$

where $\omega \geq 0$.

Classifier-Free Guidance (CFG)

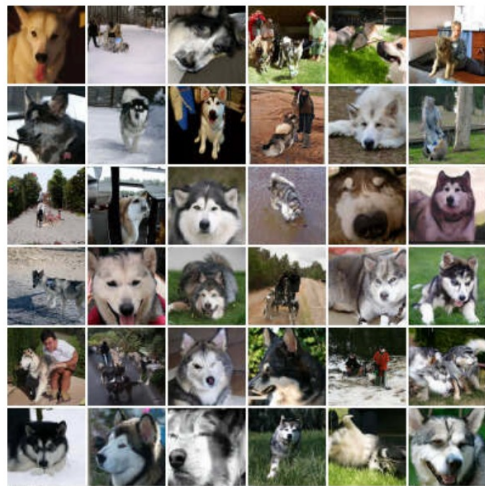
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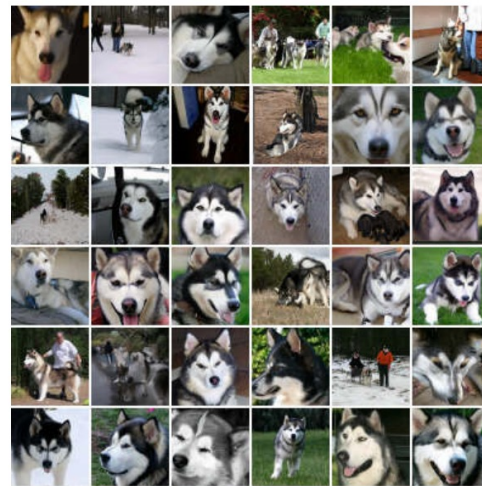
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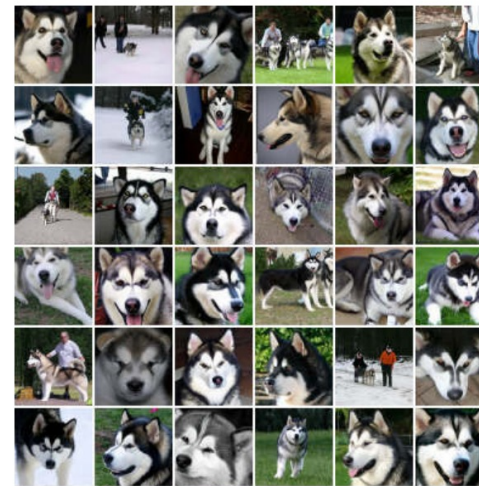
Classifier-Free Guidance (CFG)



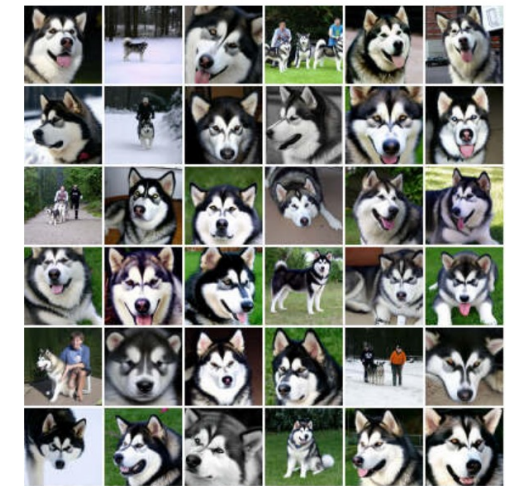
(a) $w = 0$



(b) $w = 1$



(c) $w = 2$



(d) $w = 4$

Classifier-Free Guidance (CFG)

- (+) Very easy to implement.
- (+) Versatile; can be used not only for labels but also for any additional information (e.g., text descriptions).
- (−) The noise predictor needs to be evaluated twice in the generation process.

Assignment 2

Implement CFG based on the DDIM implementation.

Wildlife



Cat



Dog

